

Identifying Behavioral Pattern of Humans Living in a Smart Home

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INTRODUCTION

Smart homes offer unparalleled comfort and efficiency, becoming increasingly accessible and popular due to advancing technology. Data plays a crucial role in the functioning of smart homes, enabling them to make informed decisions and personalize experiences for residents. The project seeks to collect data to uncover and understand these behavioral patterns. The paper also discusses the potential to revolutionize smart home technology and improve quality of life, especially for busy individuals. The paper underscores the importance of identifying behavioral trends in smart homes to reshape lifestyles and interactions with living environments in the digital age.



In this project, three methods were applied to find behavioral patterns. These are HMM, the Naïve bayes model and Decision tree. The data was collected of a person residing in an automated home. While the initial collection of this information did not occur through Excel, it was later transferred into this format for more convenient manipulation and analysis.

1	Timestamp	Action feature	Action result	Action result/num	Date	Time
2	2023-07-01T00:00:00	Back door	OFF	0	7/1/2023	0:00:00
3	2023-07-01T00:00:28	Hall	ON	100	7/1/2023	0:00:28
4	2023-07-01T00:00:56	Master Bedroom	OFF	0	7/1/2023	0:00:56
5	2023-07-01T00:01:24	Hall	ON	100	7/1/2023	0:01:24
6	2023-07-01T00:01:52	Landing	ON	100	7/1/2023	0:01:52
7	2023-07-01T00:02:20	Hall	ON	100	7/1/2023	0:02:20
8	2023-07-01T00:02:48	Master Bedroom	ON	100	7/1/2023	0:02:48
9	2023-07-01T00:03:16	Master Bedroom	ON	100	7/1/2023	0:03:16
10	2023-07-01T00:03:44	Main Kitchen	ON	100	7/1/2023	0:03:44
11	2023-07-01T00:04:12	Main Kitchen	ON	100	7/1/2023	0:04:12
12	2023-07-01T00:04:40	Master Bedroom	ON	100	7/1/2023	0:04:40
13	2023-07-01T00:05:08	Back door	OFF	0	7/1/2023	0:05:08
14	2023-07-01T00:05:36	Kettle	OFF	0	7/1/2023	0:05:36
15	2023-07-01T00:06:04	Master Bedroom	OFF	0	7/1/2023	0:06:04
16	2023-07-01T00:06:32	Hall	ON	100	7/1/2023	0:06:32

METHODS

Implementation of HMM

Hidden Markov Model (HMM) is applied to the dataset after label encoding and Timestamp column transformation. The action and result features are used to create sequence data. Hidden states for ON and OFF conditions are identified. The data is split into test and training sets, and the HMM model is trained. Parameters are printed but found to exceed probability limits, requiring normalization. Normalized parameters are then applied to the model, and a sequence is generated and decoded to demonstrate functionality.

Normalized Initial State Probabilities: [0.2691888181279986, 0.7388191898728015]
Normalized Transition Matrix: [[0.4871853496618323, 0.51281465093381678], [0.48553461967213885, 0.5144653803278612]]
Normalized Emission Matrix: [[0.048716706161988584, 0.050603397494198475, 0.05279801892498462, 0.049233854885737965, 0.0478221071822647994, 0.05681685299845565, 0.047532722170195975, 0.04815678991551185, 0.04847158904368481, 0.04768603922199561, 0.050931118410279968, 0.050880268757296239, 0.048252874202117626, 0.047890909483077976, 0.04825219339389366, 0.0497274676776453, 0.04982472336164911, 0.0508804875144077815, 0.04874563575234664, 0.049479241478189894], [0.0506136638065985, 0.04885940402961727, 0.055548511378478726, 0.04854319410786476, 0.05076966218711196, 0.051504910818463634, 0.04763681838198522, 0.047964373588659916, 0.047738888439639794, 0.04844614887085984, 0.04997660734998978, 0.0497315669499391, 0.04777562287792008, 0.04815821014753823, 0.059467102735493084, 0.04917681574374889, 0.049184180632026116, 0.0485562387755531, 0.050655234127783196, 0.04949971953875801]]

	Timestamp	Generated_Action	Generated_Result
0	2023-11-01 07:00:00	Second Bedroom	ON
1	2023-11-01 07:05:00	Back door	ON
2	2023-11-01 07:10:00	Master Bedroom	ON
3	2023-11-01 07:15:00	Fridge door	ON
4	2023-11-01 07:20:00	Kettle	OFF
...	...	...	...
95	2023-11-01 14:55:00	Second Bedroom	ON
96	2023-11-01 15:00:00	Living Room	ON
97	2023-11-01 15:05:00	Main Kitchen	ON
98	2023-11-01 15:10:00	Back door	ON
99	2023-11-01 15:15:00	Fridge door	ON

Implementation of Naïve Bayes Model

A Naive Bayes algorithm predicted "on" or "off" states based on actions, requiring label encoding. Initially, it achieved around 79.74% accuracy but performed poorly in predicting "off" states due to class imbalance.

To address this, "off" states were oversampled, and the model was retrained, resulting in roughly 63.84% accuracy. The model improved in predicting "off" states and maintained accuracy for "on" states, becoming more balanced overall. The predicted sequence was printed after decoding.

accuracy_upsampled				
0.6384625831862813				
Classification Report:				
	precision	recall	f1-score	support
0	0.31	0.61	0.41	2984
1	0.87	0.64	0.74	11742
accuracy			0.64	14726
macro avg	0.59	0.63	0.57	14726
weighted avg	0.75	0.64	0.67	14726

Fig: report

Predicted_Result	Action_feature
0	OFF Hall
1	ON Main Kitchen
2	ON Main Kitchen
3	ON Third Bedroom
4	ON Kettle
...	...
14721	ON Master Bedroom
14722	ON Kettle
14723	ON Master Bedroom
14724	OFF Fridge door
14725	ON Landing

Fig: predicted result

Implementation of Decision Tree

Categorical attributes like Action feature and Action result were converted to numeric format using the Label Encoder for training a decision tree model. The dataset was split into 80% for training and 20% for testing. The Decision Tree Classifier was trained to make predictions based on Action features.

Accuracy: 0.88				
Classification Report:				
	precision	recall	f1-score	support
0	0.51	0.06	0.11	2984
1	0.81	0.98	0.89	11742
accuracy	0.66	0.52	0.58	14726
macro avg	0.75	0.80	0.73	14726
weighted avg				

Fig: report

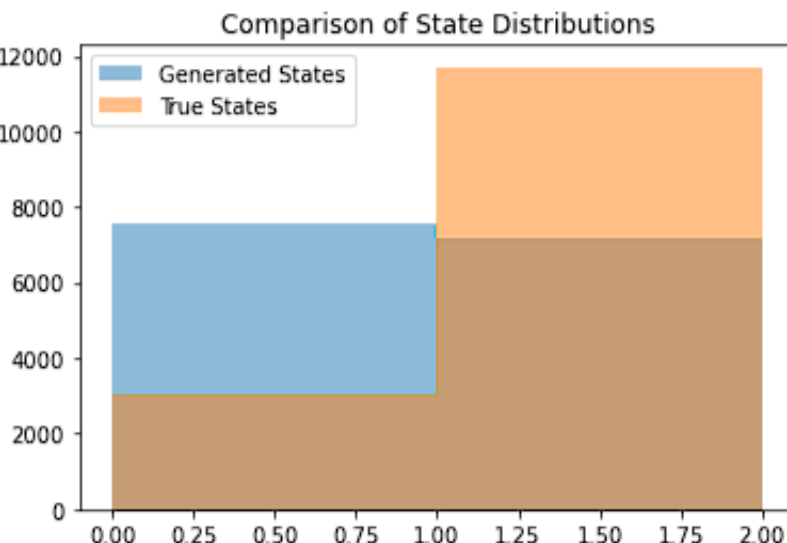
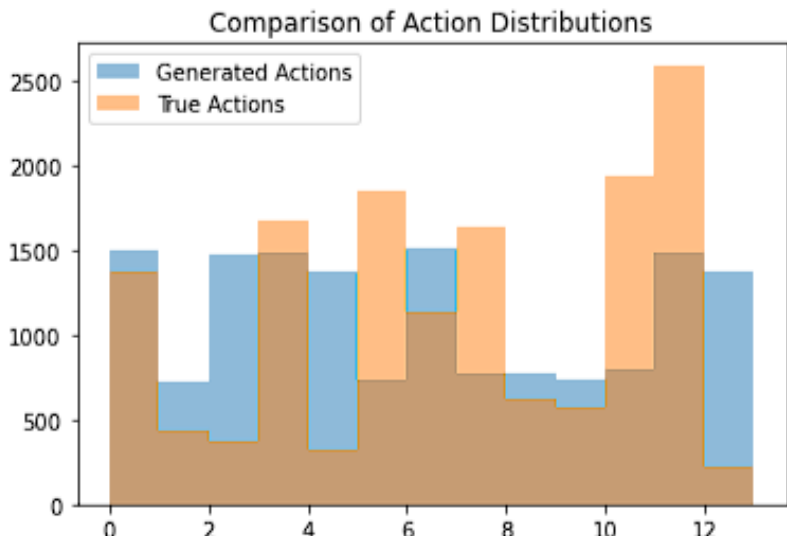
Predicted_Result	Action feature
0	ON Hall
1	ON Main Kitchen
2	ON Main Kitchen
3	ON Third Bedroom
4	ON Kettle
...	...
14721	ON Master Bedroom
14722	ON Kettle
14723	ON Master Bedroom
14724	ON Fridge door
14725	ON Landing

Fig: predicted result

EVALUATION & RESULTS

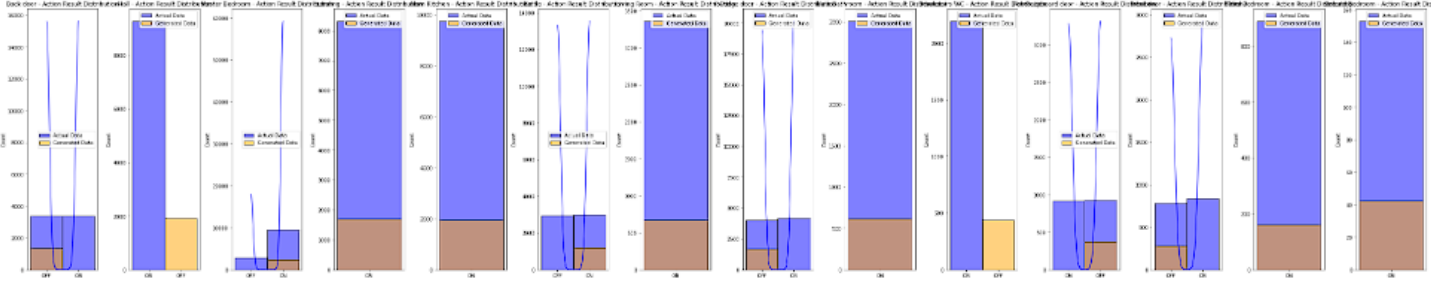
HMM

The produced comparison of state is observed to generally encapsulate the actual condition, although it does not do so flawlessly, and we can see that the generated actions are distributed in a way that is similar to the actual data in some parts and shows difference in other parts.



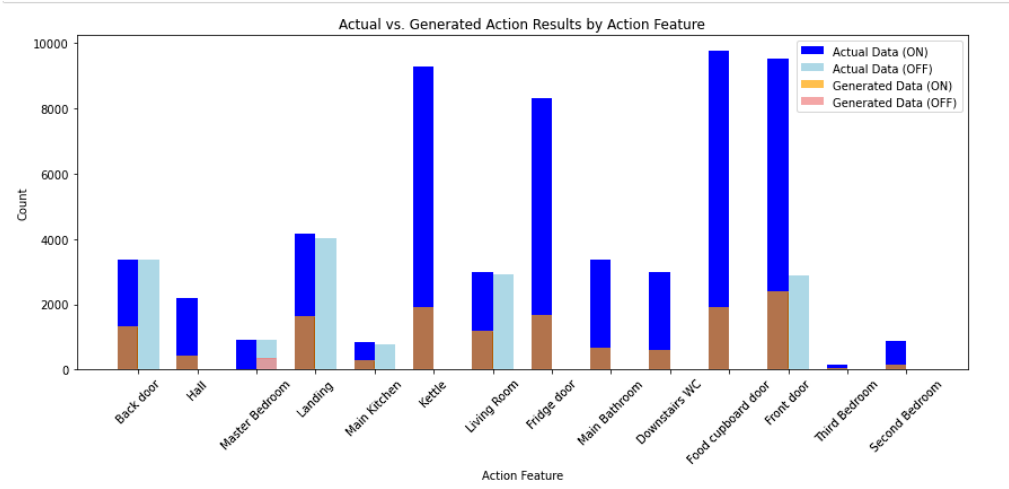
Naïve Bayes

Certain characteristics were accurately forecasted by the actual data, while others yielded divergent outcomes. For instance, features like Halls and Downstairs WC did not have any OFF state in reality but were still producing this nonexistent state. Several other attributes could only produce one particular condition without generating its counterpart.



Naïve Bayes

We see that the generated data hardly consists of the OFF state. So almost none of the feature's actual OFF data were predicted. Only the Master Bedroom's OFF data was predicted but it couldn't predict the ON state for the Master Bedroom. But other than that most of the state were predicted.



Evaluation report

Models	Accuracy	F1 Score	Recall
HMM	61.17%	0.64	0.62
Naïve Bayes	63.84%	0.67	0.64
Decision Tree	80%	0.73	0.8

CONCLUSIONS

- Decision Tree model showed 80% accuracy but variable precision between classes.
- Other models could have better results with better data selection
- There are inconsistencies in specific features.
- incorporating clustering techniques and hybrid models like HMM with neural networks could have a better result
- Other advanced machine learning practices like LSTM and Transformer models can also produce better result

This report acknowledges the potential and challenges in finding patterns analytics, offering recommendations for more accurate, adaptable, and comprehensive

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